CONTROL SYSTEMS

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Faculty team

RRC:	ECC:
Dr. Anuradha M – Anchor	Prof. Muralidhar S
Dr. Aswini N, Dr. Rashmi N.U	**EEE:**
Dr. Anagha K	Prof. K.R.Pushpa

MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Block Diagram Reduction Techniques

Transfer Functions

There are major two ways of obtaining a transfer function for the control system.

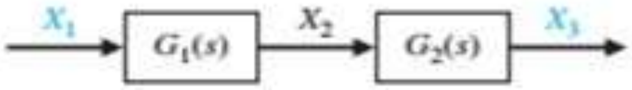
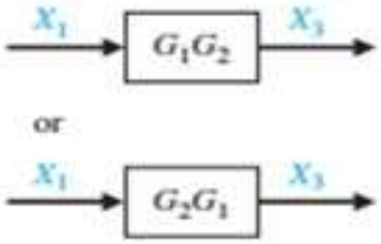
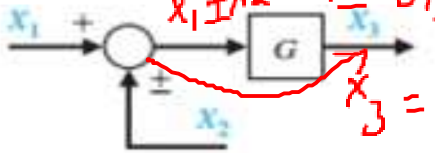
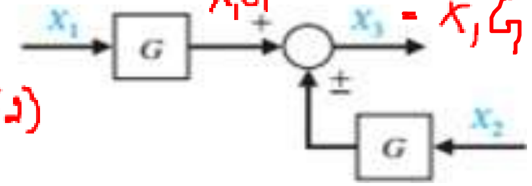
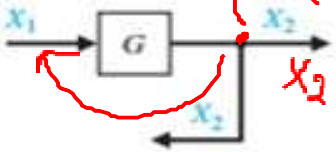
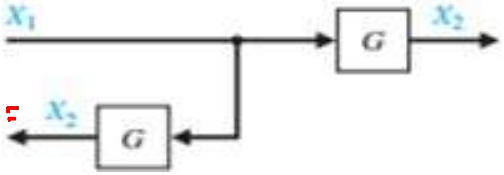
- **Block Diagram Reduction**
- **Signal Flow Graphs**
- **State space Model**

Block Diagram Reduction Technique:

- It is not convenient to derive a complete transfer function for a complex control system.
- Therefore the transfer function of each element of a control system is represented by a block diagram.
- Block diagram reduction techniques are applied to obtain the desired transfer function.
- Block diagram gives a pictorial representation of a control system.

MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Transfer Functions using Block Diagram

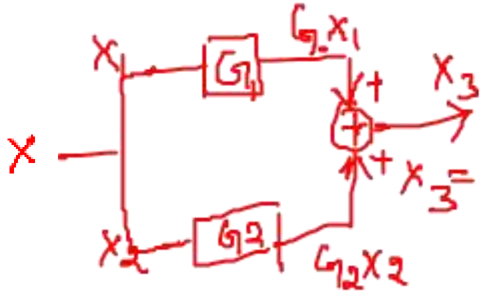
Transformation	Original Diagram	Equivalent Diagram
1. Combining blocks in cascade	 $X_2 = G_1 X_1$ $x_3 = G_2 X_2$	
2. Moving a summing point behind a block	$X_1 \pm X_2 = G_2 G_1 X_1$  $X_3 = G(x_1 \pm x_2)$	$X_1 G$  $X_3 = X_1 G \pm G X_2$
3. Moving a pickoff point ahead of a block	pick-off or tap-off  $X_2 = G X_1$	 $G X_1 = X_2$

Transfer Functions using Block Diagram

4. Moving a pickoff point behind a block

5. Moving a summing point ahead of a block

⑦ 2 blocks in parallel

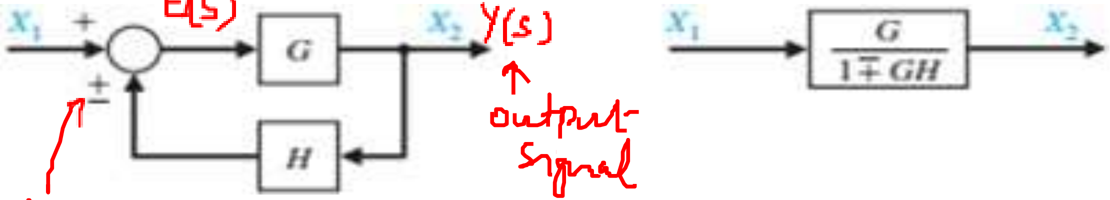


$X_3 = G_1X_1 + G_2X_2$

$= (G_1 + G_2)X$

Reference i/P $R(s)$ → $E(s)$ error signal or actuating signal

6. Eliminating a feedback loop



$Y(s)$ ↑ output signal

feedback

$E(s) = X_1 \pm HX_2$

$X_2 = G\,E(s)$

$E(s) = X_1 \pm HGE(s)$

$E(s)(1 \mp HG) = X_1$

G – Forward path T F

H – feedback path T.F

$E(s) = \dfrac{X_1}{1 \mp GH}$

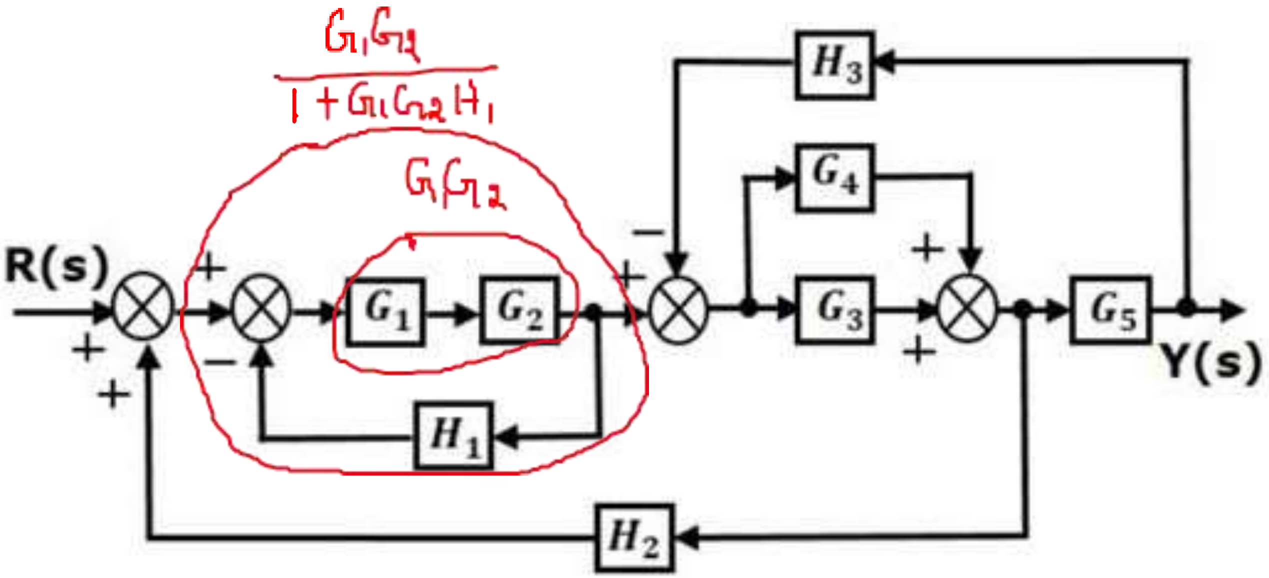
$X_2 = G\,E(s)$

$= \dfrac{G X_1}{1 \mp GH}$

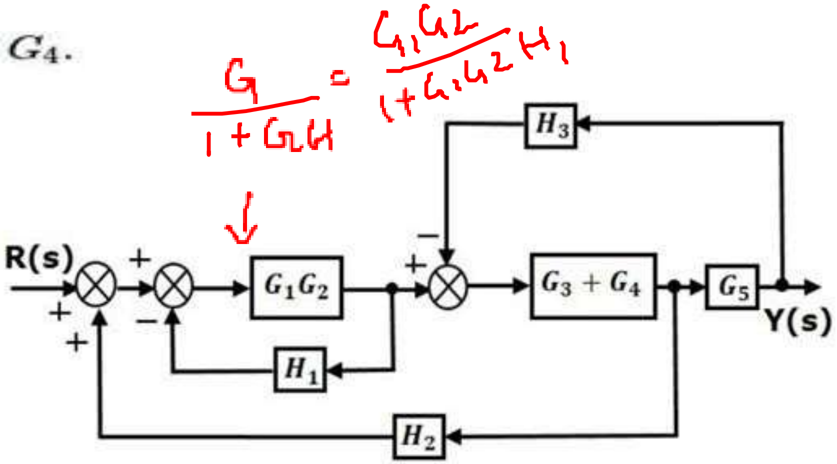
$\dfrac{X_2}{X_1} = \dfrac{G}{1 \mp GH}$

MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Transfer Functions using Block Diagram – Example 1



Step 1 – Use Rule 1 for blocks G_1 and G_2. Use Rule 2 for blocks G_3 and G_4. The modified block diagram is shown in the following figure.



MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Transfer Functions using Block Diagram - Example

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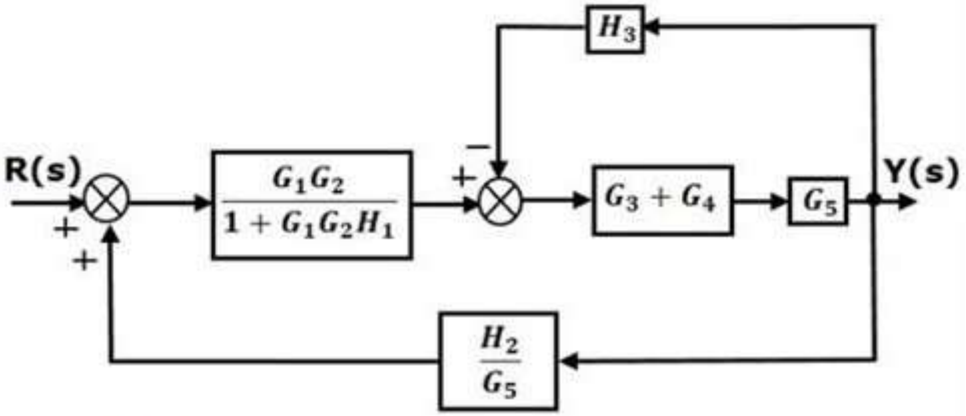
Step 2 – Use Rule 3 for blocks G_1G_2 and H_1. Use Rule 4 for shifting take-off point after the block G_5. The modified block diagram is shown in the following figure.



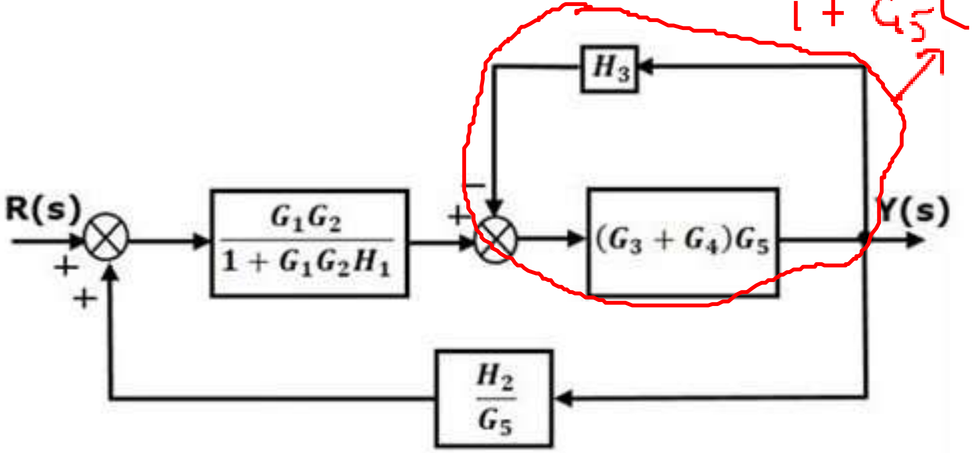
$$G = G_5(G_3+G_4)$$

$$H = H_3$$

$$= \frac{G}{1+GH}$$

$$= \frac{G_5(G_3+G_4)}{1+G_5(G_3+G_4)H_3}$$

Step 3 – Use Rule 1 for blocks $(G_3 + G_4)$ and G_5. The modified block diagram is shown in the following figure.



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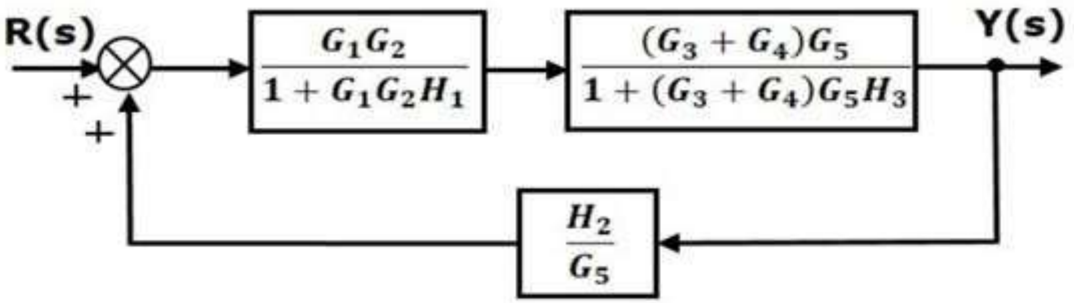
Transfer Functions using Block Diagram - Example

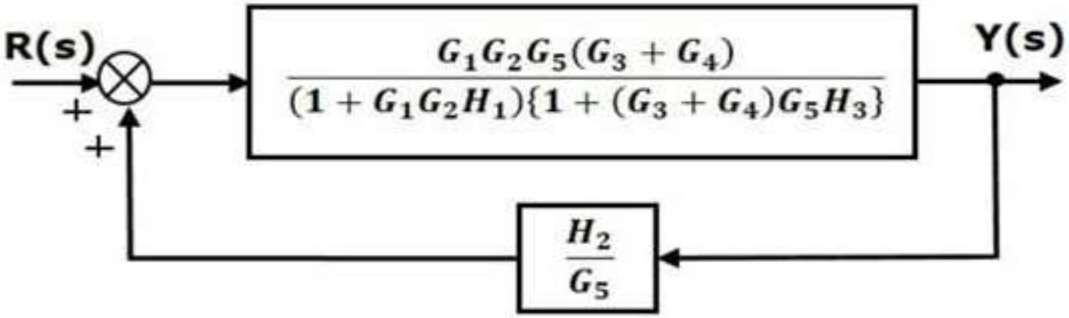
Step 4 – Use Rule 3 for blocks $(G_3 + G_4)G_5$ and H_3. The modified block diagram is shown in the following figure.



Step 5 – Use Rule 1 for blocks connected in series. The modified block diagram is shown in the following figure.

$$\frac{Y(s)}{R(s)} = \frac{G}{1-GH}$$

G



H

Step 6 – Use Rule 3 for blocks connected in feedback loop. The modified block diagram is shown in the following figure. This is the simplified block diagram.

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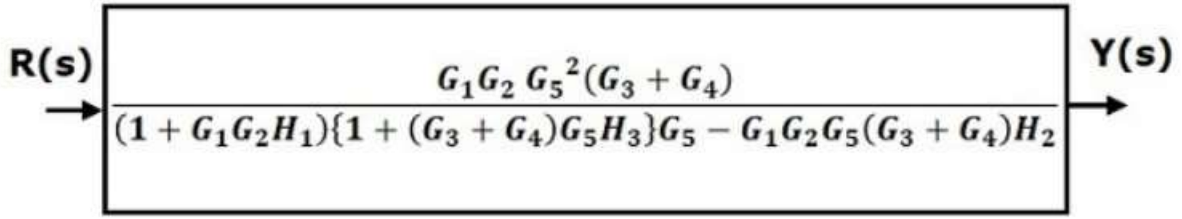
Transfer Functions using Block Diagram - Example

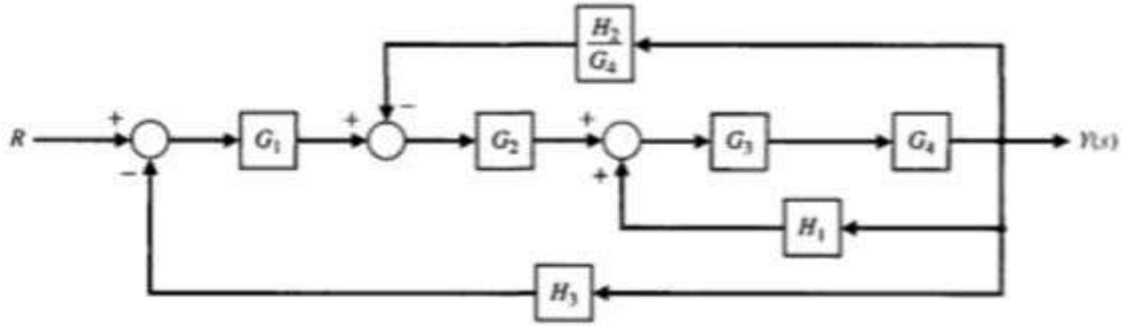


Therefore, the transfer function of the system is

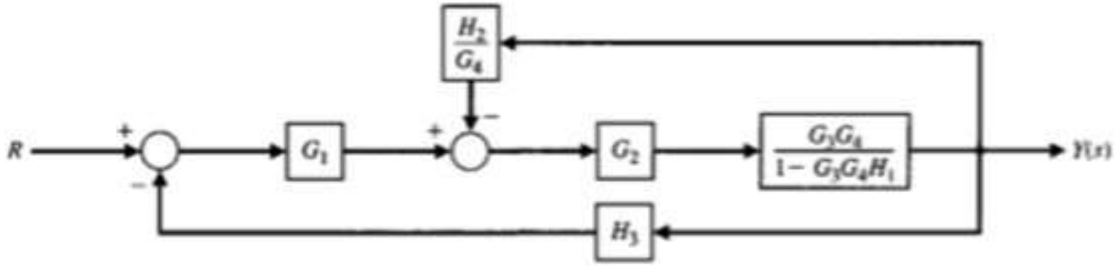
$$\frac{Y(s)}{R(s)} = \frac{G_1G_2G_5^2(G_3+G_4)}{(1+G_1G_2H_1)\{1+(G_3+G_4)G_5H_3\}G_5-G_1G_2G_5(G_3+G_4)H_2}$$

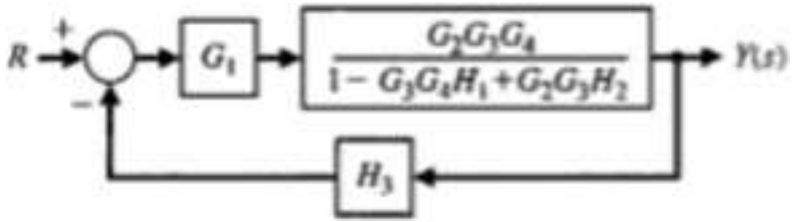
MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

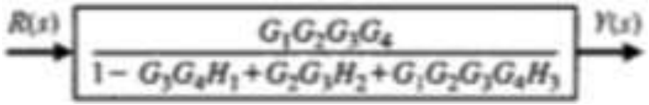
Transfer Functions using Block Diagram – Example 2



$$\frac{G}{1+GH}$$

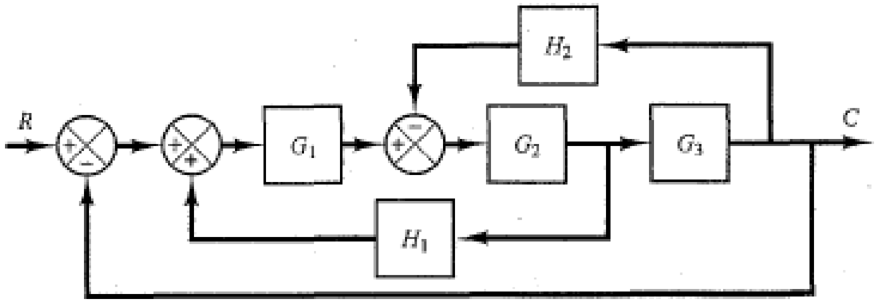


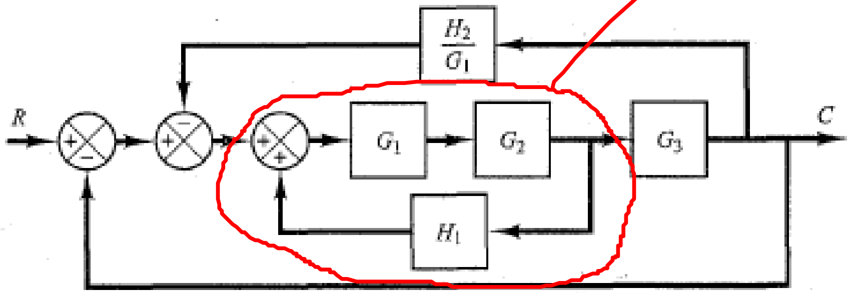




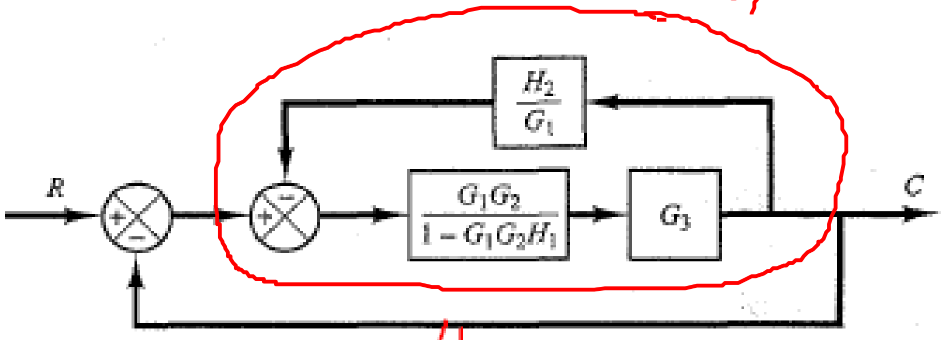
MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Transfer Functions using Block Diagram – Example 3

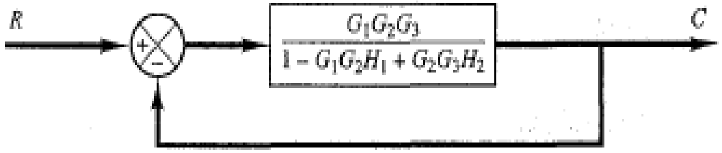




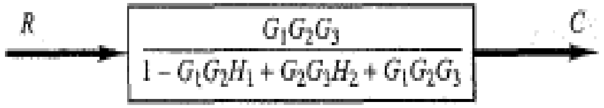
$$\frac{G_1G_2}{1 - G_1G_2H_1}$$



$$\frac{\dfrac{G_1G_2G_3}{1 - G_1G_2H_1}}{1 + \dfrac{\cancel{G_1}G_2G_3}{1 - G_1G_2H_1} \cdot \dfrac{H_2}{\cancel{G_1}}}$$



$$\frac{C}{R} = \frac{\dfrac{G_1G_2G_3}{1 - G_1G_2H_1 + G_2G_3H_2}}{1 + G_1G_2G_3 / 1 - G_1G_2H_1 + G_2G_3H_2} = \frac{G_1G_2G_3}{1 - G_1G_2H_1 + G_2G_3H_2}$$



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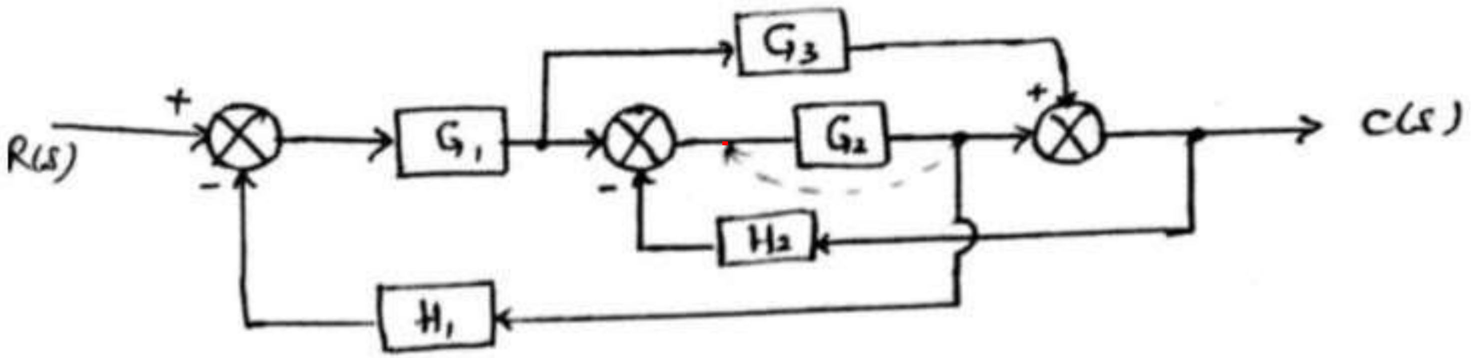
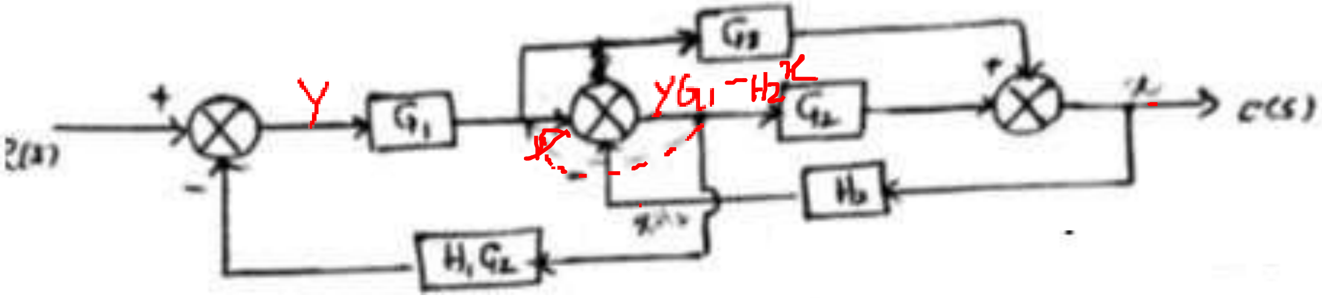
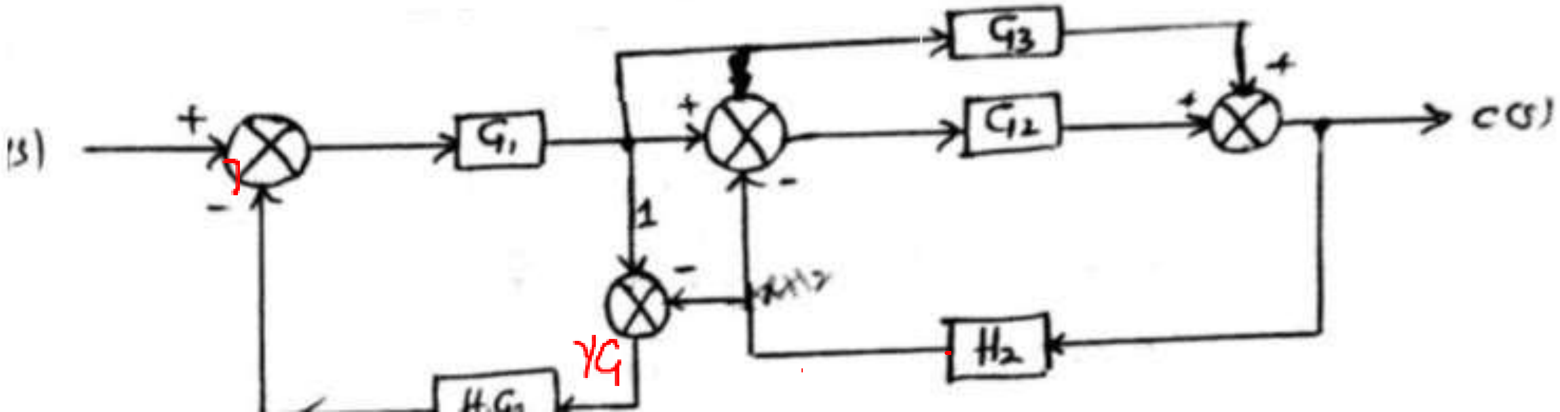
Transfer Functions using Block Diagram – Example 4







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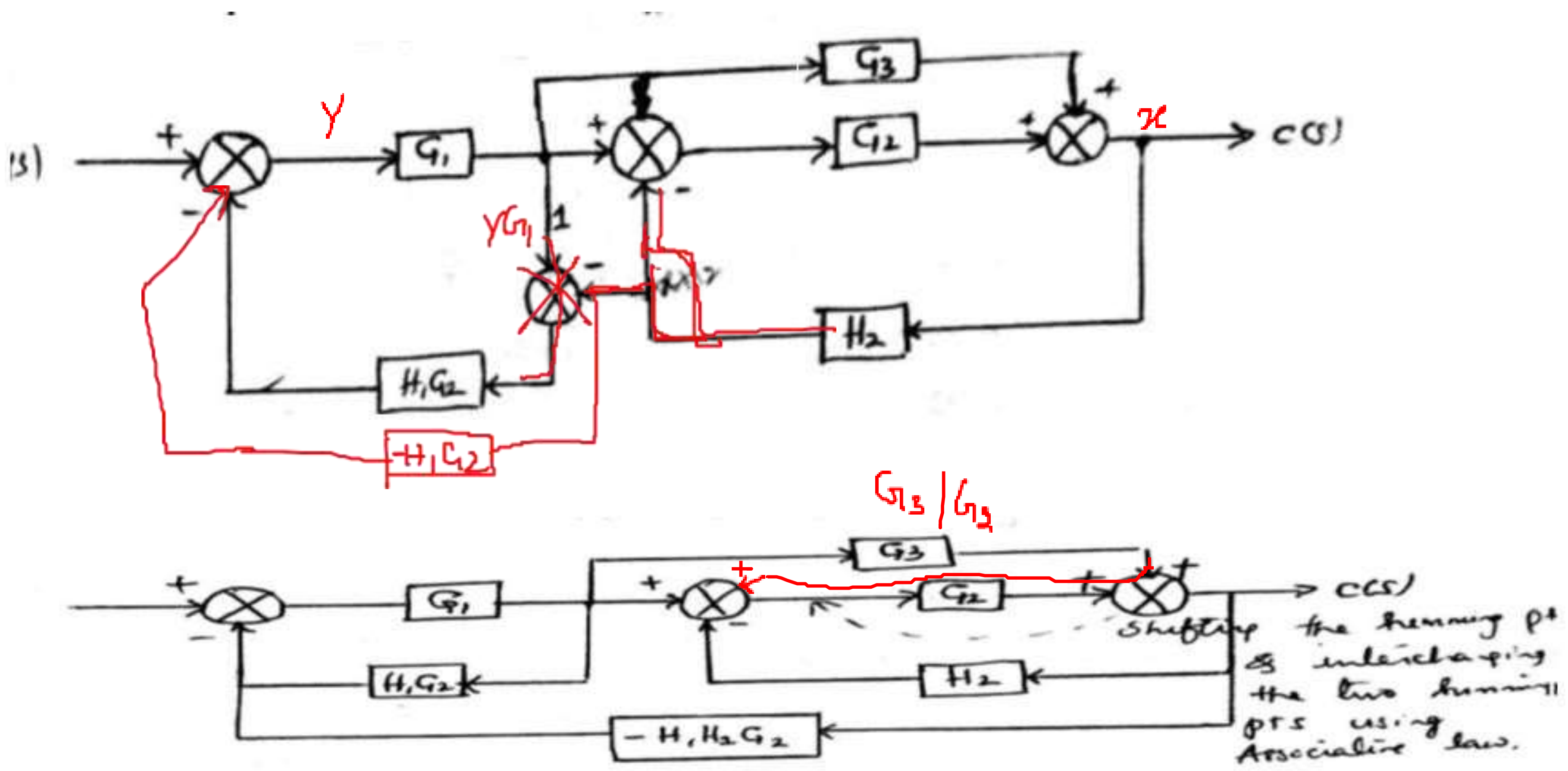
Transfer Functions using Block Diagram – Example 4



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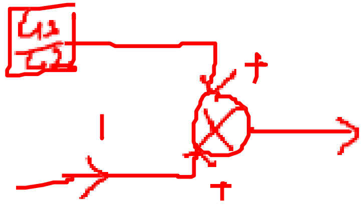
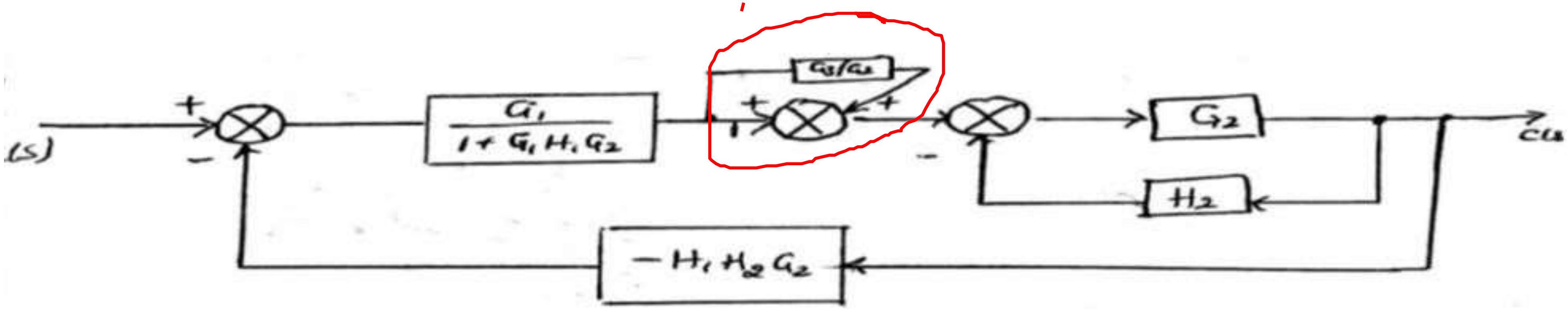
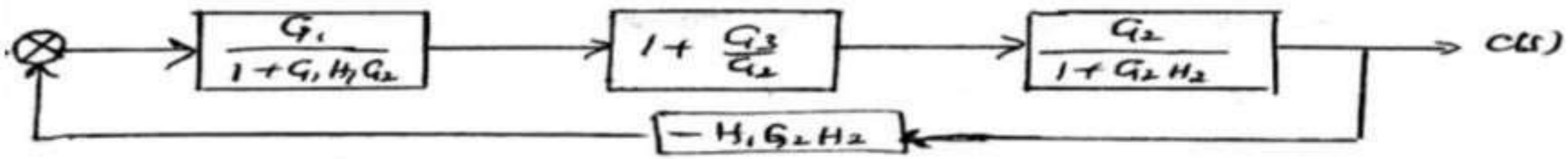
Transfer Functions using Block Diagram – Example 4







On simplification.

$$\boxed{\frac{C(s)}{R(s)} = \frac{G_1 (G_2 + G_3)}{1 + G_2 H_2 + H_1 G_1 G_2 - G_1 G_2 G_3 H_1 H_2}}$$

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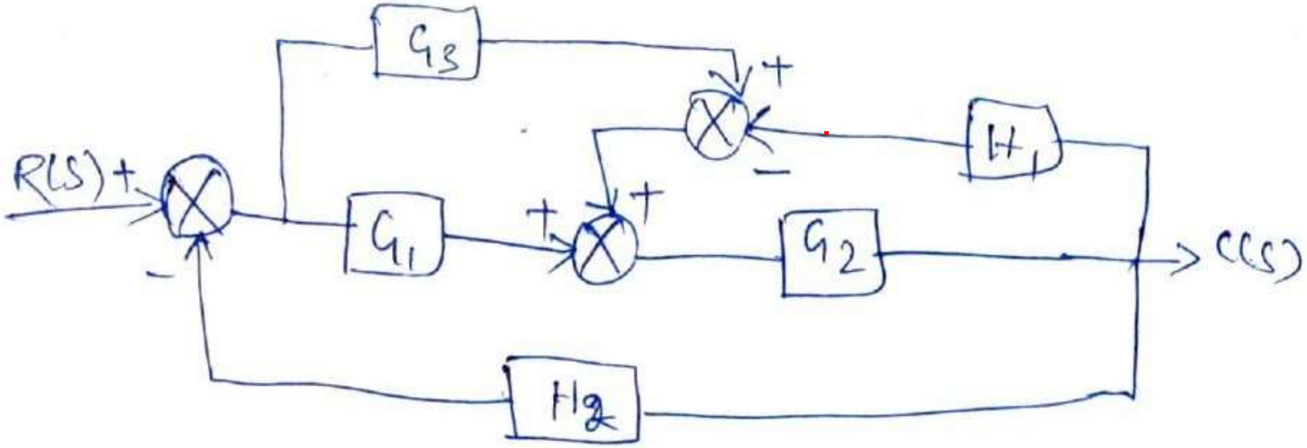
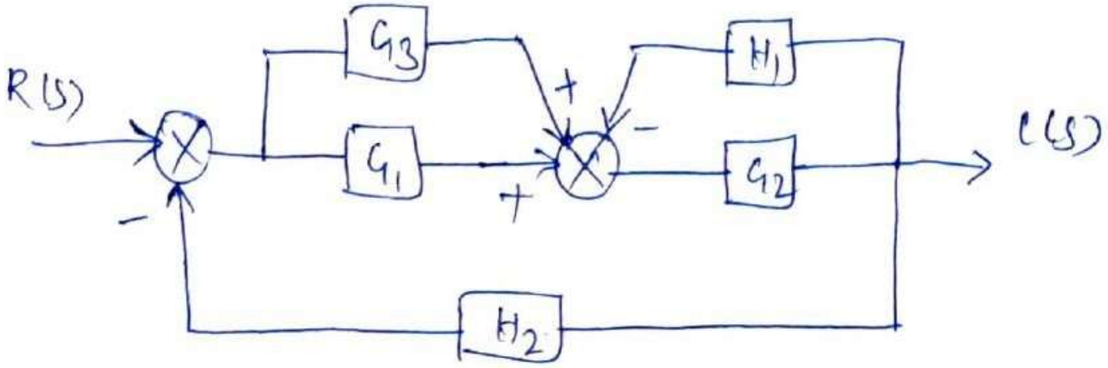
Transfer Functions using Block Diagram – Example 5





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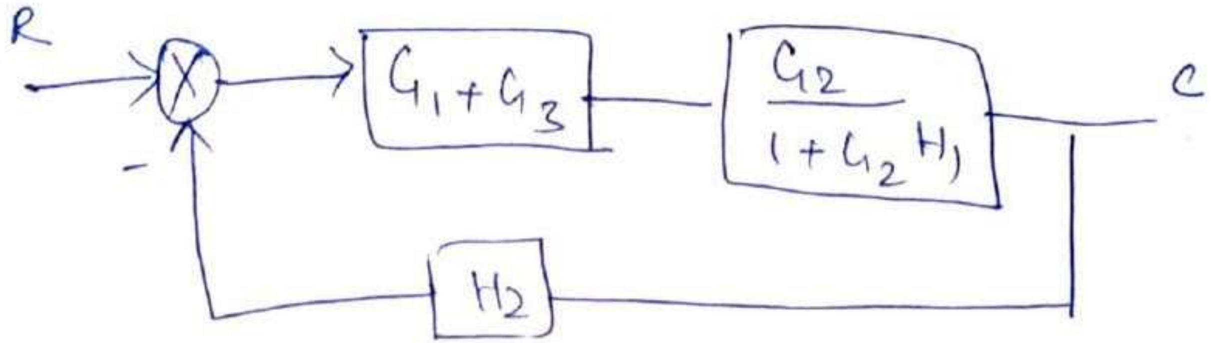
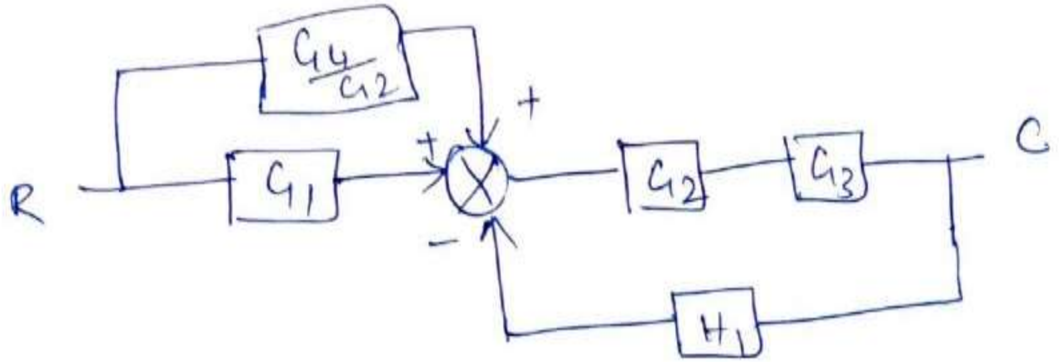
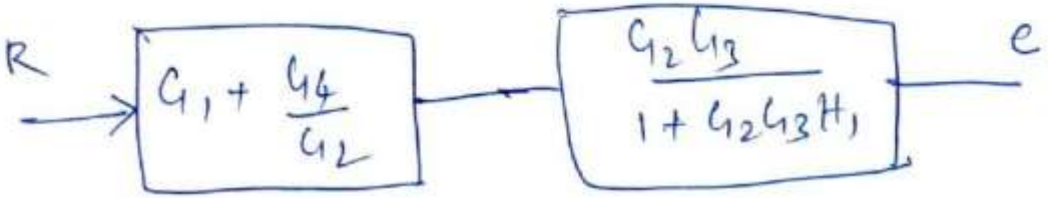
Transfer Functions using Block Diagram – Example 5



$$\frac{C}{R} = \frac{G_1G_2 + G_2G_3}{1 + G_2H_1 + G_1G_2H_2 + G_2G_3H_3}$$

Transfer Functions using Block Diagram – Example 6



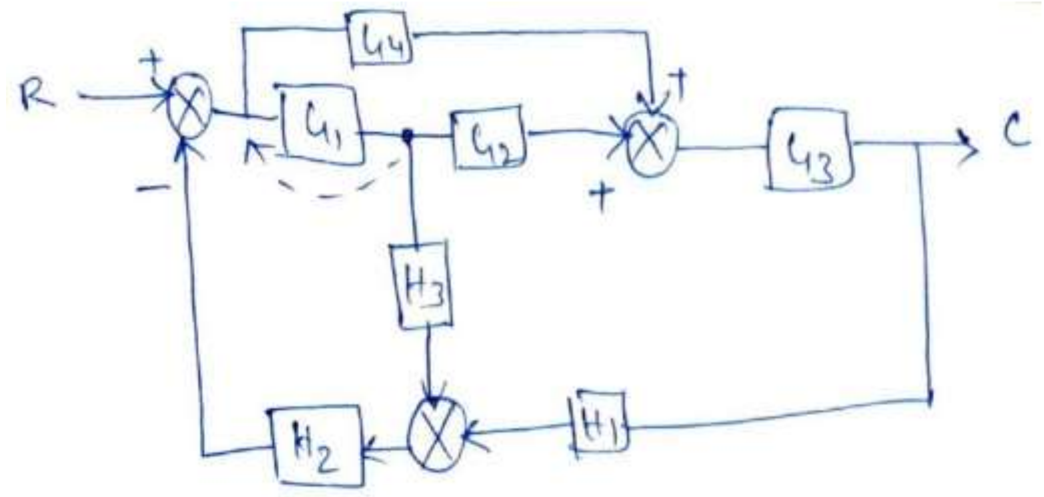


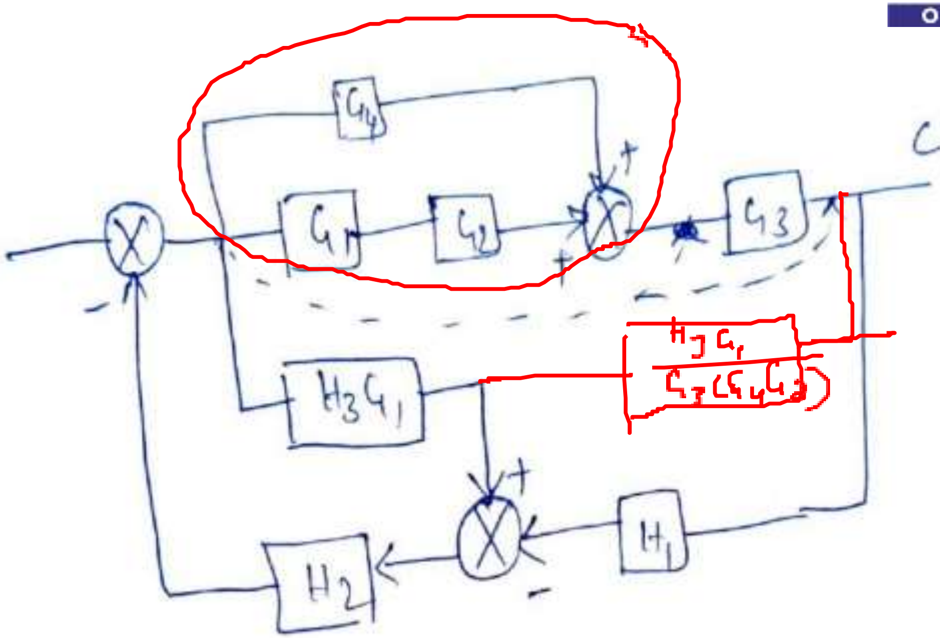
$$\frac{C}{R} = \frac{(G_1G_2 + G_4)\,G_3}{1 + G_2G_3H_1}$$

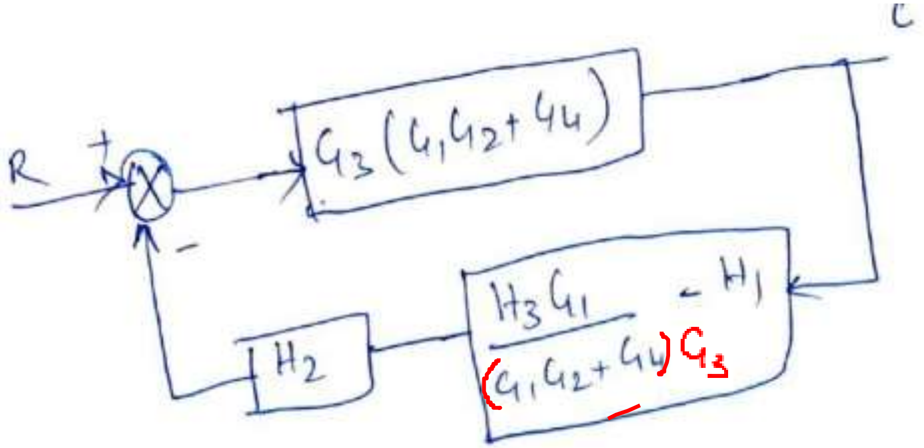
MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Transfer Functions using Block Diagram – Example 7

$(G_4 + G_1 G_2) G_3$







$$\frac{C}{R} = \frac{G_3(G_1G_2+G_4)}{1 + G_1H_2H_3 - H_1H_2G_3(G_1G_2+G_4)}$$

THANK YOU

Dr. Anuradha M
Department of Electronics and Communication Engineering

anuradha@pes.edu

+91 80 2672 1983   Extn: 774